| Vidyalankar Institute of Technology<br>Accredited A+ by NAAC | Department of Computer Engineering<br>A.Y-2026-27 |
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| Semester | B.E. Semester VII – CMPN |
| Subject | DevOps |
| Subject Professor In-charge | Prof. Snehal Andhare |
| Assisting Teachers | Prof. Snehal Andhare |
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| Grade and Subject Teacher's Signature | | |

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| Experiment Number | 12 |
| Experiment Title | Luggage Complaint Portal: Jenkins-Docker Continuous Deployment |
| Objectives (Skill Set / Knowledge Tested / Imparted) | To extend the Jenkins pipeline to build a versioned Docker image, publish it to a registry and deploy a fresh container automatically after the tests pass. |
| Program | **1. Local registry and Docker access for Jenkins**<br>`docker run -d --name registry --restart unless-stopped -p 5000:5000 registry:2`<br>`sudo usermod -aG docker jenkins`<br>`sudo systemctl restart jenkins`<br>**2. Docker stages added to the Jenkinsfile**<br>`environment { IMAGE = 'localhost:5000/luggage-portal' TAG = "${env.BUILD_NUMBER}" // version tag = build number }`<br>`stage('Docker Build') { steps { sh 'docker build -t $IMAGE:$TAG -t $IMAGE:latest .' } }`<br>`stage('Docker Push') { steps { sh 'docker push $IMAGE:$TAG && docker push $IMAGE:latest' } }`<br>`stage('Docker Deploy') { steps { sh 'docker rm -f portal \|\| true docker run -d --name portal --restart unless-stopped -p 8083:8080 $IMAGE:$TAG' } }` |

### 3. Versioned images in the registry

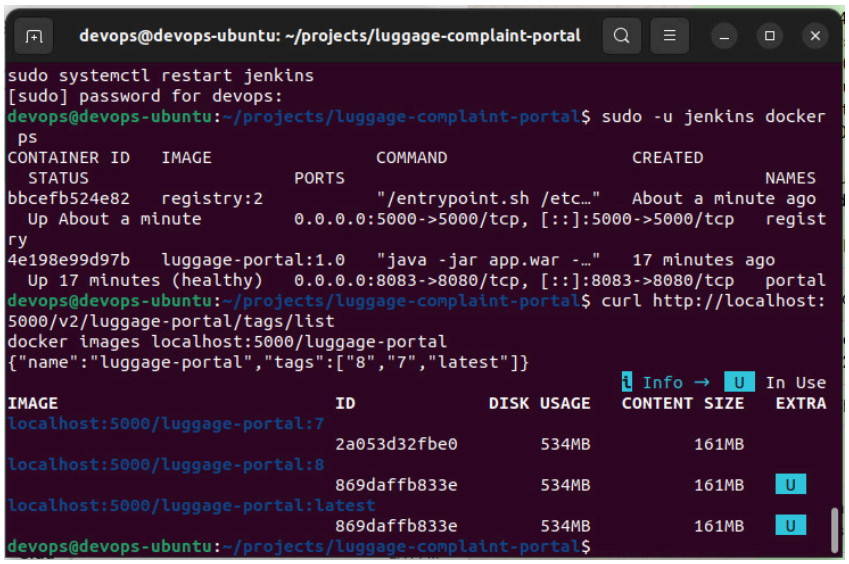



*Figure: Registry tags ["8","7","latest"], docker images and the running portal container*

### 4. End-to-end pipeline run

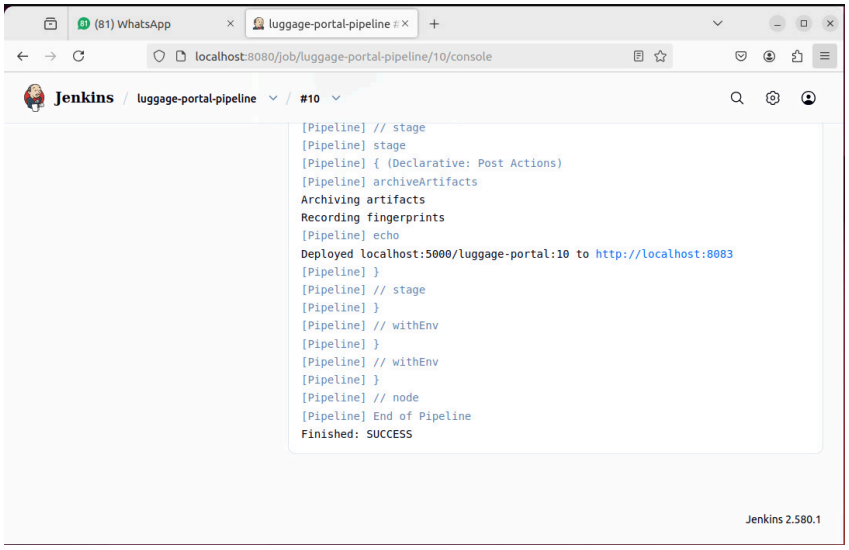
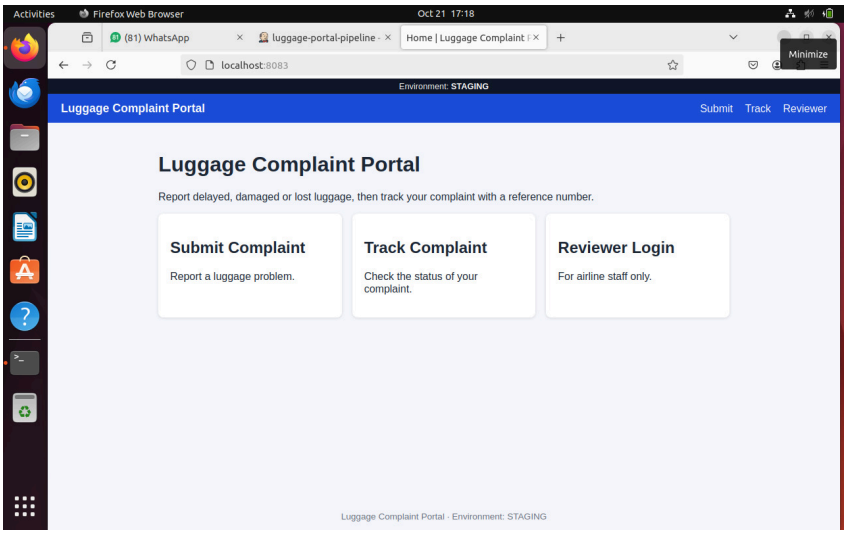



*Figure: Build #10: Deployed localhost:5000/luggage-portal:10 to http://localhost:8083, Finished: SUCCESS*

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| | **5. Fresh container serving the new build**<br>  <br>*Figure: Application on port 8083 showing Environment: STAGING* |
| Conclusion | The pipeline builds a versioned image, pushes it to a local registry and deploys a fresh container automatically after the tests pass, from commit to container. |